

S-Band RHCP Patch Antenna

Simulation Report

Rogers RO4350 Substrate · 2.42 GHz · HFSS 2025 R2.4

1. Document Control

Version	Date	Description
V0	2026-06-07	Initial document

2. Simulation Environment

Item	Value	Notes
Simulation Software	Ansys HFSS 2025 R2.4	
Solver Type	Frequency Domain FEM	Adaptive mesh refinement
Frequency Sweep Type	Discrete	Discrete used for far-field AR vs freq
Adaptive Convergence	$\Delta S < 0.02$	
Radiation Boundary	Radiation (PML equivalent)	Air box dimensions: 300*300*200 mm
Port Type	Coaxial Wave Port	Terminal S-parameter solution Copper pin, Teflon based dielectric Inner radius = 0.635 mm Outer radius = 2.065 mm

3. Design Parameters

3.1 Substrate

Parameter	Value	Notes
Material	Rogers RO4350	
Relative Permittivity (ϵ_r)	3.66	
Loss Tangent ($\tan \delta$)	0.004	
Substrate Thickness (h)	1.52 mm	Thickest option offered by JLCPCB as of V0
Substrate Dimensions	100 × 100 mm	Ground plane / substrate extent
Copper Cladding Thickness	34.8 μm (1 oz)	Option from JLCPCB as of V0

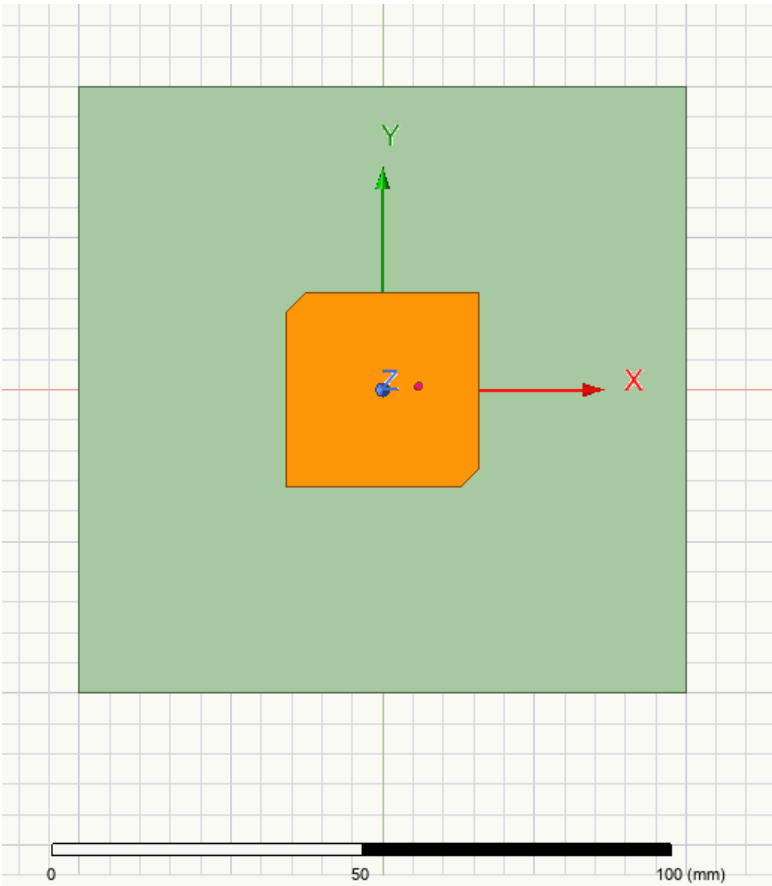
3.2 Patch Geometry

Parameter	Value	Notes
Patch Shape	Square	Corner-truncated for CP
Patch Side Length (L)	31.8 × 31.8 mm	
Corner Truncation (c)	3 mm	Leg of removed right-isosceles triangle
Truncated Corners	Top-Left & Bottom-Right	Produces RHCP
Intact Corners	Top-Right & Bottom-Left	

3.3 Feed

Parameter	Value	Notes
Feed Type	Coaxial Probe	SMA centre pin, 1.27 mm dia.
Feed X-Offset (feedx)	5.9 mm	From patch centre, +X direction
Feed Y-Offset (feedy)	0.6 mm	From patch centre, +Y direction
Feed Impedance	50 Ω	
Feed Axis	Off-diagonal	feedx ≠ feedy ensures both modes excited

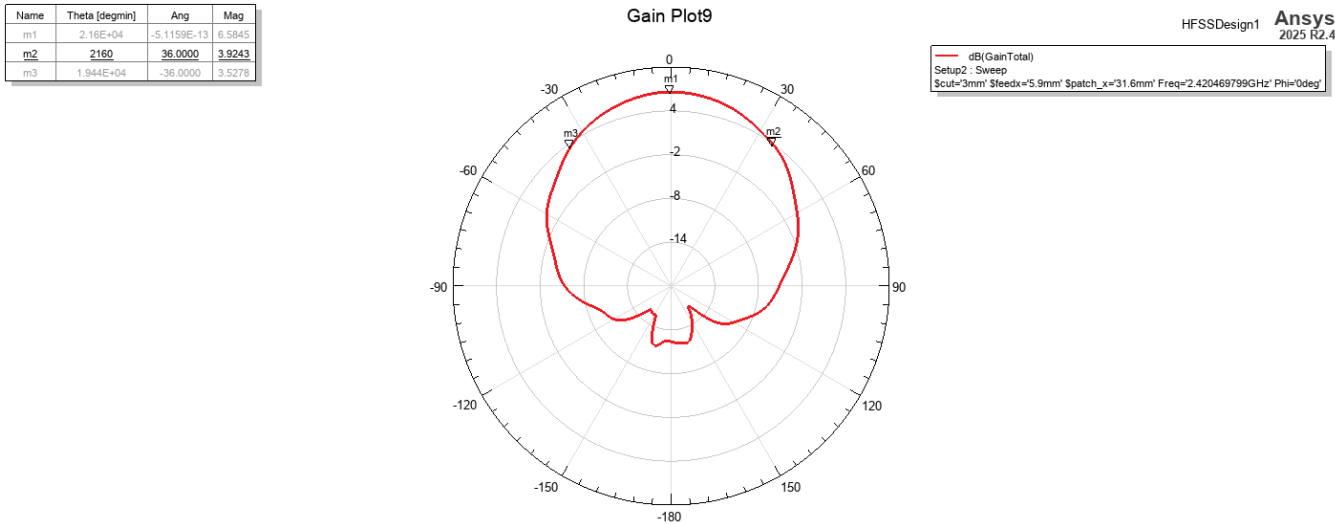
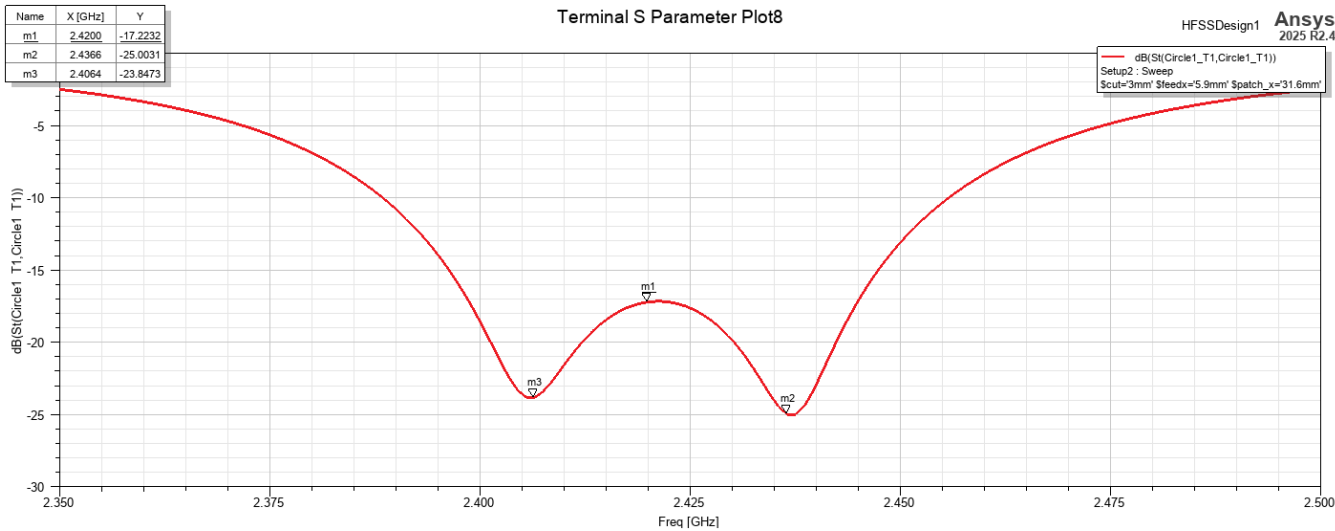
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4. Simulation Results

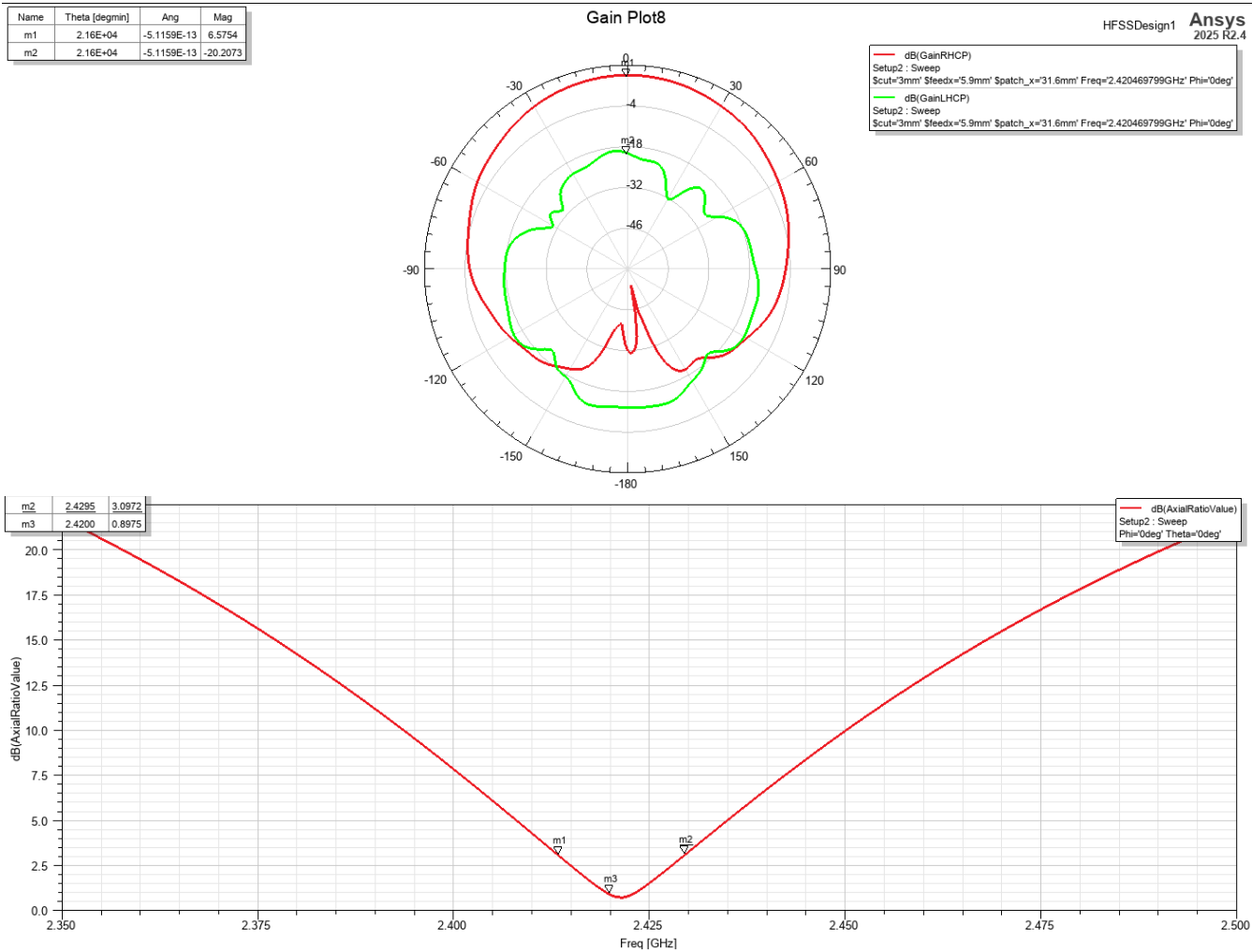
4.1 Impedance — S11

Parameter	Result	Notes
Centre Frequency (f0)	2.42 GHz	Demonstration purpose, will need to retune post frequency allocation
Lower Resonance (f1)	2.4064 GHz	First split mode dip
Upper Resonance (f2)	2.4366 GHz	Second split mode dip
Mode Split (f2 – f1)	30.2 MHz	Target $\approx f_0 / Q = 60$ MHz
S11 at f1	-23.85 dB	
S11 at f2	-25.00 dB	
Dip Depth Balance	1.15 dB	
S11 at f0	-17.22 dB	Midpoint between dips — expected for CP
Simulated Q Factor	≈ 80.1	$Q = f_0 / \text{split} = 2420 / 30.2$

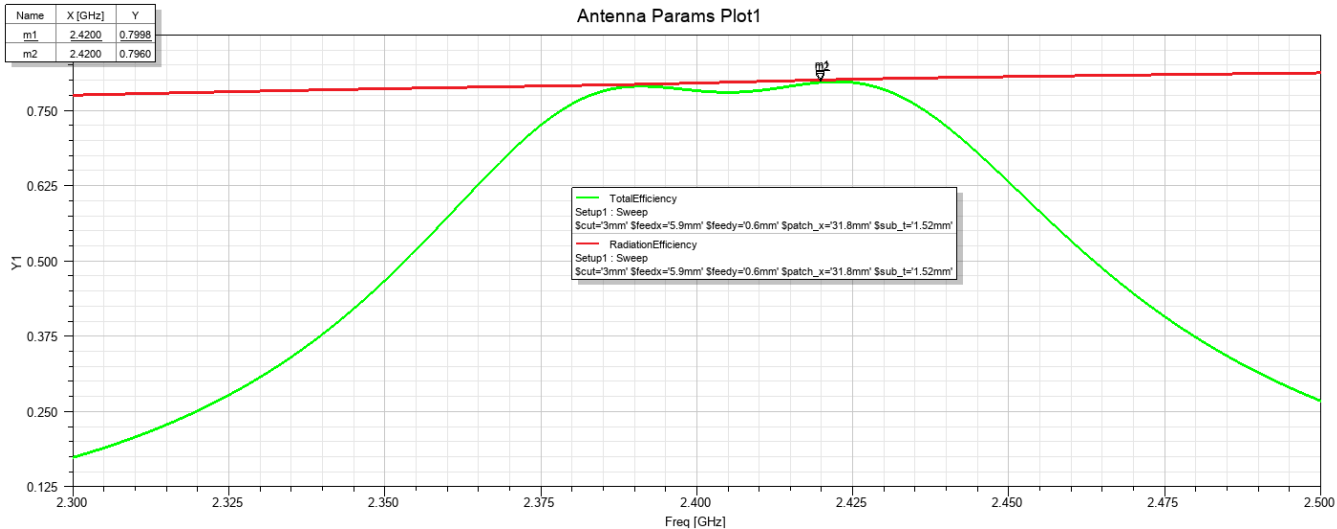
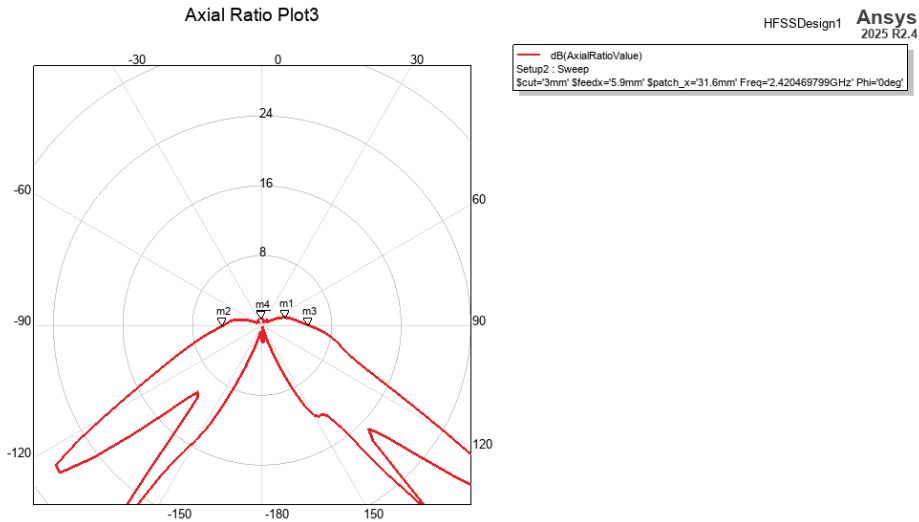


4.2 Radiation Performance

Parameter	Result	Notes
Polarisation Sense	RHCP	Confirmed from GainRHCP >> GainLHCP
GainRHCP at Boresight	+6.58 dB	$\theta = 0^\circ, \varphi = 0^\circ, f = 2.42 \text{ GHz}$
GainLHCP at Boresight	-20.2 dB	Cross-polarisation component
Cross-Pol Discrimination (XPD)	26.78 dB	GainRHCP – GainLHCP at boresight
Axial Ratio at Boresight	0.80 dB	< 3 dB threshold
AR at $\theta = +90^\circ$ (horizon)	5.42 dB	Expected degradation off-boresight
AR at $\theta = -90^\circ$ (horizon)	4.46 dB	
AR Beamwidth (< 3 dB)	$\approx 16 \text{ MHz}$	Determine sufficiency post frequency allocation
AR 3dB Bandwidth	$\sim 144 \text{ deg}$	
Peak Gain	6.58	
Half-Power Beamwidth	72 deg	
Radiation Efficiency	79.98 %	Extract from HFSS efficiency report
Total Efficiency	79.60 %	System efficiency



Name	Theta [degmin]	Ang	Mag
m1	4380	72.0000	2.9160
m2	1.62E+04	-90.0000	4.4554
m3	5400	90.0000	5.4166
m4	2.16E+04	-5.1159E-13	0.7962



5. Link Budget Context

Parameter	Value	Notes
Application	CubeSat S-Band Downlink	Qubesat Mission
Target Frequency	2.42 GHz	Need retuning post frequency allocation
Orbit Altitude	LEO (TBD km)	Likely 400 km
Max Doppler Shift	±61 kHz	At 7.5 km/s orbital velocity
AR BW vs Doppler Margin	16 MHz / 0.12 MHz ≈ 133	Ample margin
Required Polarisation	RHCP	Standard ground station convention

6. Next Steps

#	Action	Owner	Status
1	Fabrication Review	TBD	Pending